Better Understanding Brighter Future

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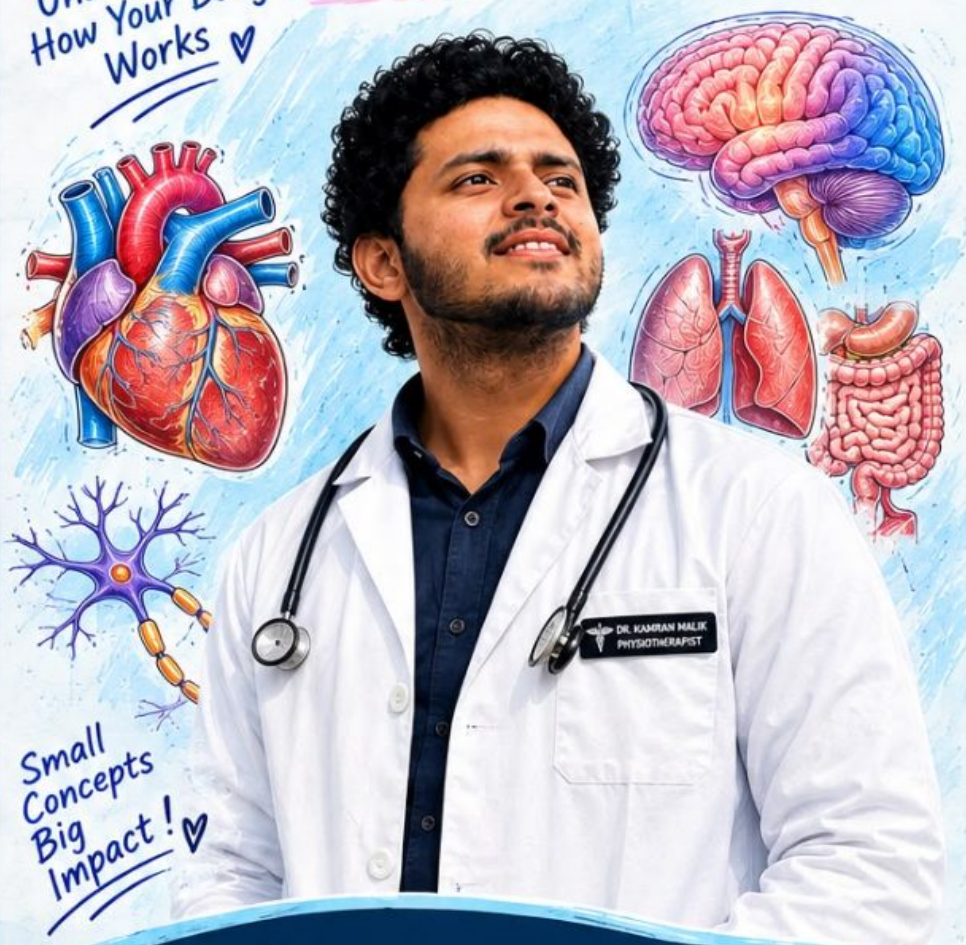
HUMAN PHYSIOLOGY

THE COMPLETE STUDY GUIDE

FOR MBBS | BHMS | B.Sc. NURSING | BDS | BPT

Understand How Your Body Works

Easy • Quick • Exam-Oriented Notes



Small Concepts Big Impact!

KEY AREAS

- ✓ Cell Physiology
- ✓ Membrane Physiology
- ✓ Nerve & Muscle
- ✓ Cardiovascular System
- ✓ Respiratory System
- ✓ Renal System
- ✓ GIT & Liver
- ✓ Endocrine System
- ✓ Reproductive System
- ✓ Special Senses
- ✓ Homeostasis

WHY THIS BOOK?

- High-Yield Points
- Concept Clarity
- Diagrams & Flowcharts
- Previous Year Exam Focus
- Easy to Revise

CELL PHYSIOLOGY | NERVE & MUSCLE | CARDIOVASCULAR | RESPIRATORY | RENAL | ENDOCRINE

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Better Physiology Better Care

Healthy Body Healthy Tomorrow

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Human Physiology

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

HUMAN PHYSIOLOGY

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Human Physiology

Topic – 11

Cardiovascular System

– Introduction & Organization –

1. Introduction :-

- Cardiovascular system (CVS) is a highly efficient system responsible for continuous circulation of blood throughout the body.
- It supplies oxygen and nutrients to tissues and removes carbon dioxide and metabolic waste products.
- It consists of heart, blood vessels and blood.
- Works in close coordination with respiratory, renal, endocrine and nervous systems.

2. Functions of Cardiovascular System :-

a) Transport of oxygen, nutrients, hormones and wastes.
b) Regulation of body temperature.
c) Maintenance of fluid and electrolyte balance.
d) Protection (immune cells, clotting factors).
e) Helps in maintaining pH and acid-base balance.

3. Components of Cardiovascular System :-

a) **Heart** – muscular pump.
b) **Blood vessels** – arteries, veins and capillaries.
c) **Blood** – transport medium.

4. Organization of Cardiovascular System :-

a) **Heart** :-
- Located in the mediastinum (between lungs).
- Enclosed in a double-walled sac called pericardium.
- Divided into 4 chambers – Right atrium, Right ventricle, Left atrium, Left ventricle.
- Acts as a double pump – right side (pulmonary circulation), left side (systemic circulation).

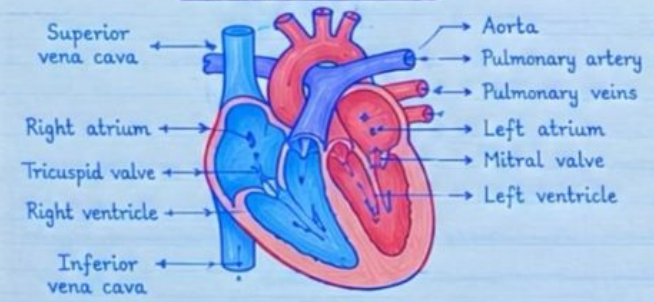
b) **Blood Vessels** :-
- **Arteries** – carry blood away from heart (usually oxygenated).
- **Veins** – carry blood towards heart (usually deoxygenated).
- **Capillaries** – site of exchange of gases, nutrients and wastes.

c) **Blood** :-
- Plasma – liquid part (55%).
- Formed elements – RBCs, WBCs, Platelets (45%).

5. Structure of the Heart :-

- Four chambers – RA, RV, LA, LV.
- Valves – Tricuspid, Pulmonary, Mitral (Bicuspid), Aortic.
- Coronary circulation – supplies oxygen and nutrients to heart muscle.
- Cardiac muscle – involuntary, myogenic.
- Conducting system – SA node, AV node, Bundle of His, Right & Left bundle branches, Purkinje fibres.

Structure of Heart



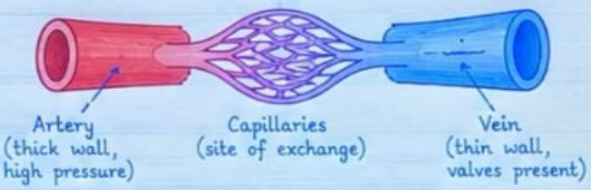
6. Blood Vessels – Types & Organization :-

a) **Arteries** :-
- Carry blood away from heart.
- Thick, elastic and muscular walls.
- High pressure, no valves (except at origin).
- Types – elastic, muscular, arterioles.

b) **Capillaries** :-
- Site of exchange of gases, nutrients and wastes.
- Very thin walls (one cell thick).
- Connect arterioles to venules.

c) **Veins** :-
- Carry blood towards heart.
- Thin walls, less muscle and elasticity,
- Contain valves (prevent backflow).
- Function as blood reservoir.

Types of Blood Vessels

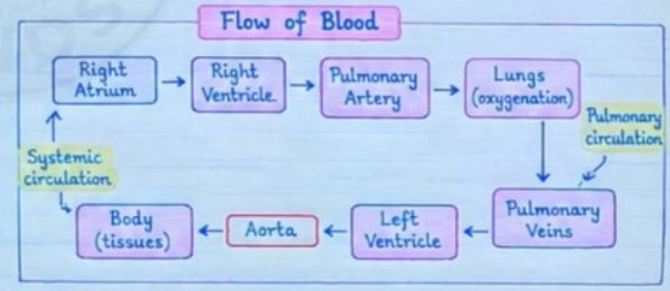


7. Circulation of Blood :-

a) **Pulmonary Circulation** (Right Heart → Lungs → Left Heart) :
1. Deoxygenated blood from superior & inferior vena cava enters right atrium.
2. Passes through tricuspid valve → right ventricle.
3. Goes through pulmonary artery to lungs (gets oxygenated).
4. Returns via pulmonary veins to left atrium.

b) **Systemic Circulation** (Left Heart → Body → Right Heart) :
1. Oxygenated blood from left atrium → mitral valve → left ventricle
2. Ejected through aorta → systemic arteries → capillaries (oxygen & nutrient exchange).
3. Returns via systemic veins → superior & inferior vena cava → right atrium.

Flow of Blood



8. Summary :-

- Heart = 4 chambers, 4 valves, 2 pumps.
- Blood vessels = arteries, veins, capillaries.
- Blood = plasma + RBCs + WBCs + platelets.
- Pulmonary circulation = right heart → lungs → left heart.
- Systemic circulation = left heart → body → right heart.
- Together, they maintain life by continuous circulation.

Key Points to Remember :-

★ Right side = deoxygenated blood (pulmonary).
★ Left side = oxygenated blood (systemic).
★ Arteries (except pulmonary) carry oxygenated blood.
★ Veins (except pulmonary) carry deoxygenated blood.
★ Capillaries = exchange.

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THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

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